

- General Math**
- Identities**
- $|\mathbf{v} - \mathbf{u}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2 - 2|\mathbf{u}||\mathbf{v}|\cos\theta$
 - $\sec^2\theta = 1 + \tan^2\theta, \quad \csc^2\theta = 1 + \cot^2\theta$
 - $\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$
 - $\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$
 - $\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$
 - $\cos^2\theta = \frac{1}{2}(1 + \cos(2\theta))$
- Determinants**

- The determinant of a matrix is the “scaling factor” of the hypercube when the linear mapping is applied. In particular, in $\mathbb{R}^3$, the 3×3 matrix gives the **volume of the parallelepiped** (or its negative).
- Determinant is **unchanged** by rearranging the rows/columns.
 - Computed by **cofactor expansion**.
 - $|A| = |A^T|$

Coordinate Systems

In the **Cartesian coordinate system**, we use the **right hand rule** to identify the x, y, z axes (Index towards the x , middle towards the y , thumb towards the z).

Distance between two points is given by:

$$\left|\overrightarrow{P_1P_2}\right| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}.$$

We change to the other coordinate systems with:

$$r = \sqrt{x^2 + y^2}, \quad \tan\theta = \frac{y}{x}, \quad z = z.$$

$$\rho = \sqrt{x^2 + y^2 + z^2}, \quad \cos\phi = \frac{z}{\rho}, \quad \tan\theta = \frac{y}{x}.$$

In the **Cylindrical coordinate system**, we represent each point P by (r, θ, z) , such that $r \geq 0$ is the length the projection of $\overrightarrow{OP}$ on the xy -plane and θ its angle with the x -axis. We change to the other coordinate systems with:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z.$$

$$\rho = \sqrt{r^2 + z^2}, \quad \cos\phi = \frac{z}{\rho}, \quad \theta = \theta.$$

In the **Spherical coordinate system**, we represent each point P by (ρ, ϕ, θ) such that ρ is the distance from the origin, $\phi \in [0, \pi]$ is the angle between $\overrightarrow{OP}$ and the z -axis. We change to the other coordinate systems with:

$$r = \rho\sin\phi, \quad \theta = \theta, \quad z = \rho\cos\phi.$$

$$x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi.$$

Vector Spaces

A set V equipped with addition and scalar multiplication on V satisfying the following properties is a **vector space** over $\mathbb{F}$:

- Commutative. $\forall u, v \in V$,
$$u + v = v + u$$
- Associative. $\forall u, v, w \in V$ and $\forall a, b \in \mathbb{F}$,
 $(u + v) + w = u + (v + w)$ and $(ab)w = a(bw)$.
- Additive Identity. $0 \in V$. $\forall v \in V$,
$$v + 0 = v$$
- Additive Inverse. $\forall v \in V, \exists ! w \in V$,
$$v + w = 0.$$
- Multiplicative Identity. $1 \in \mathbb{F}$. $\forall v \in V$,
$$1v = v$$
- Distributive Properties. $\forall u, v \in V$ and $\forall a, b \in \mathbb{F}$,
 $a(u + v) = au + av$ and $(a + b)v = av + bv$.

$\forall k \in \mathbb{N}, \mathbb{R}^k$ is a vector space over $\mathbb{F}$.

Given a point P , the position vector $\overrightarrow{OP} = \langle x_P, y_P, z_P \rangle$

and its length $\left|\overrightarrow{OP}\right| = \sqrt{x_P^2 + y_P^2 + z_P^2}.$

For any nonzero $\mathbf{v}$, $\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|}$ is a unit vector in the direction of $\mathbf{v}$.

Vector Products

The **dot product** of two vectors $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^3$ is defined:

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 = |\mathbf{u}||\mathbf{v}|\cos\theta$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

- $\mathbf{u}, \mathbf{v}$ are **orthogonal** $\iff \mathbf{u} \cdot \mathbf{v} = 0$.
- $\forall \mathbf{v}, (\mathbf{0}, \mathbf{v})$ are orthogonal.
- The **projection** of $\mathbf{u}$ in the direction of $\mathbf{v}$ is defined:

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} \right) \hat{\mathbf{v}}.$$

The scalar component can be negative, when $\theta \in \left] \frac{\pi}{2}, \pi \right]$

The **cross product** of two vectors $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ and $\mathbf{c} = \langle c_1, c_2, c_3 \rangle$ is defined:

$$\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= |\mathbf{u}||\mathbf{v}|(\sin\theta)\mathbf{n}.$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$ and $\mathbf{n}$ is the unit vector in the direction according to the right hand rule.

$$\bullet \quad \forall \mathbf{a} \in \mathbb{R}^3, \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a}$$

- Cross product is defined in $\mathbb{R}^2$ by taking them as vectors on the xy -plane of $\mathbb{R}^3$.
- $\mathbf{u}, \mathbf{v}$ are **parallel** $\iff \mathbf{u} \times \mathbf{v} = \mathbf{0}$
- The cross product is the **area of the parallelogram**.
- Properties** of the cross product:
 - $(r\mathbf{u}) \times (s\mathbf{v}) = rs(\mathbf{u} \times \mathbf{v})$
 - $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$
 - $(\mathbf{v} + \mathbf{w}) \times \mathbf{u} = (\mathbf{v} \times \mathbf{u}) + (\mathbf{w} \times \mathbf{u})$
 - $\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$
 - $\mathbf{0} \times \mathbf{u} = \mathbf{0}$
 - $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin\theta$

2 “Pre-Calculus”

Lines

For a line L in $\mathbb{R}^3$ through a point $P_0(x_0, y_0, z_0)$ and parallel to nonzero $\mathbf{v} \in \mathbb{R}^3$, we have the **vector equation**:

$$\mathbf{r} = \langle x_0, y_0, z_0 \rangle + t\mathbf{v} \quad \text{for } t \in \mathbb{R}.$$

Equivalently, we have the standard **parametrisation** of the line:

$$x = x_0 + tv_1, \quad y = y_0 + tv_2, \quad z = z_0 + tv_3 \quad \text{for } t \in \mathbb{R}.$$

Given a point S , its **distance to the line** L is given by: $\frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|}$

Planes

For a plane M in $\mathbb{R}^3$ containing a point $P_0(x_0, y_0, z_0)$ and perpendicular to nonzero $\mathbf{n} = \langle A, B, C \rangle$, we have the **vector and component equations**:

$$\mathbf{n} \cdot \overrightarrow{PP_0} = 0.$$

$$Ax + By + Cz = Ax_0 + By_0 + Cz_0.$$

To find the **distance from a point** from a point to a plane, find a vector from the plane to the point $\mathbf{v}$, and take the dot product of $\mathbf{v}$ with $\mathbf{n}$ to find the scalar multiple of $\mathbf{n}$.

$$\text{Distance} = \left| \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} \right|.$$

Cylinders

Given a curve $z = f(y)$ on the yz -plane, “moved” along the x -axis, we have points $(x, y, f(y))$ on the **cylinder generated**

by the curve, where x is any real number. The surface has the equation $z = f(y)$. The other planes and axes can also be used.

Quadric Surfaces

A **quadric surface** is a surface with an equation of the form:

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$$

Using the taylor expansion of a Given a quadric surface, we may obtain either of the equations:

$$Ax^2 + By^2 + Cz^2 + D = 0.$$

$$Ax^2 + By^2 + Cz = 0.$$

by a rotation about a principal axis, followed by a translation. WLOG, we may assume $A > 0, b \neq 0$ Using this equation, we may then **plot** the quadric surface. We have the equations:

- Sphere** of radius a with center (x_0, y_0, z_0) :
$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2.$$
$$r^2 + z^2 = a^2.$$
$$\rho = a.$$
$$\rho = \cos\theta.$$

- Ellipsoid**:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

- Elliptic Paraboloid**:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}.$$

- Elliptical Cone**:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}.$$

$$\phi = \alpha.$$

- Hyperboloid (One Sheet)**:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1.$$

- Hyperboloid (Two Sheets)**:

$$\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

- Hyperbolic Paraboloid**:

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}, \quad c > 0.$$

3 Vector Calculus

We call functions $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^3$ a **vector function** or a **vector-valued function**. We may write $\mathbf{r}$ as:

$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$. We may interpret $\mathbf{r}(t)$ as a point $P = (f(t), g(t), h(t))$ moving in space over time t . We form a **curve** in space with its path.

Limits

For vector function $\mathbf{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$ and vector $\mathbf{L}$, $\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{L} \iff \forall \epsilon > 0 \exists \delta > 0$ such that
$$\forall t, 0 < |t - t_0| < \delta \implies |\mathbf{r}(t) - \mathbf{L}| < \epsilon.$$

It can be shown that the limit of a vector function comprises the limits of its coordinates:

$$\lim_{t \rightarrow t_0} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow t_0} f(t), \lim_{t \rightarrow t_0} g(t), \lim_{t \rightarrow t_0} h(t) \right\rangle.$$

The vector function is **continuous at a point** $t = t_0$ in its domain if

$$\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{r}(t)$$

and it is a **continuous function** $\iff$ it is continuous at every point in its domain.

Differentiation

We define the derivative of a vector function as

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}.$$

$$\mathbf{r}'(t) = \frac{df}{dt}\hat{i} + \frac{dg}{dt}\hat{j} + \frac{dh}{dt}\hat{k}.$$

The derivative does not exist if the limit does not exist. We may interpret a nonzero derivative as a **tangent vector** to the curve at the given t . We then have the **tangent line** that which is spanned by $\mathbf{r}'(t)$.

If $\forall t, \mathbf{r}'(t)$ exists, $\mathbf{r}(t)$ is **differentiable**. If $\mathbf{r}(t)$ is differentiable (continuous) and $\forall t, \mathbf{r}'(t) \neq \mathbf{0}$, the curve is **smooth**.

We may also interpret derivatives as the **velocity** of the point moving in space over time. That is, $\mathbf{v} = \frac{d\mathbf{r}}{dt}$, **speed** = $|\mathbf{v}|$, **acceleration**

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$, and the **direction of motion** at time t is $\frac{\mathbf{v}}{|\mathbf{v}|}$

Properties of derivatives:

- $\frac{d}{dt} \mathbf{C} = \mathbf{0}$
- $\frac{d}{dt} [\mathbf{c}\mathbf{u}(t)] = \mathbf{c}\mathbf{u}'(t)$
- $\frac{d}{dt} [\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$
- $\frac{d}{dt} [\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$
- $\frac{d}{dt} [\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$
- $\frac{d}{dt} [\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

Integration

We define the **indefinite integral** $\int \mathbf{r} \, dt$ the set of all anti-derivatives of $\mathbf{r}$.

$$\int \mathbf{r} \, dt = \left\langle \int f(t) \, dt, \int g(t) \, dt, \int h(t) \, dt \right\rangle.$$

The **definite integral** is defined, given integrable components over the interval $[a, b]$: $\int_a^b \mathbf{r} \, dt =$

$$\left(\int_a^b f(t) \, dt \right) \hat{i} + \left(\int_a^b g(t) \, dt \right) \hat{j} + \left(\int_a^b h(t) \, dt \right) \hat{k}$$

Application: Projectile Motion

We have the following results from application of vector calculus on **projectile motion**, thrown at an angle α above the x -axis with initial velocity $\mathbf{v}_0$, and $v_0 = |\mathbf{v}_0|$:

- Horizontal displacement $x = tv_0\cos\alpha$
- Vertical acceleration $y'' = -g$
- Vertical velocity $\frac{dy}{dt} = -gt + C_1 = v_0\sin\alpha - gt$
- Vertical displacement $y = tv_0\sin\alpha - \frac{1}{2}gt^2 + C_2$
- Displacement $\mathbf{r} = \langle rv_0\cos\alpha, tv_0\sin\alpha - \frac{1}{2}gt^2 \rangle$
- Velocity $\mathbf{v} = \langle v_0\cos\alpha, v_0\sin\alpha - gt \rangle$
- Maximum height at $y' = 0, tmax = \frac{v_0\sin\alpha}{g}$
- $y = 0$ at $t = 0$ or $t = \frac{2v_0\sin\alpha}{g}$

Arc Lengths

To motivate the derivation of arc length, we consider a short straight segment of the curve

$$\Delta L \approx \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

$$= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \Delta t$$

Taking the sum of the small arc segments, we obtain the length of the smooth curve $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle, \quad a \leq t \leq b$

as t increases from $t = a$ to $t = b$ is:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

We call L the **arc length** of the curve.

Considering a function $s(t) = \int_{t_0}^t |\mathbf{v}(t)| dt$ the arc length from t_0 to t , we have the **unit tangent vector** of the smooth curve

$$\mathbf{T}(t) = \frac{\mathbf{v}(t)}{|\mathbf{v}(t)|} = \frac{d\mathbf{r}(s)}{ds} = \frac{d\mathbf{r}(t)/dt}{ds/dt}.$$

Curvature

As a measure of the amount of “curving”, we define the **curvature** of a smooth curve:

$$\kappa = \left| \frac{d\mathbf{T}(s)}{ds} \right| = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right|.$$

That is, the rate of change of direction of the unit tangent vector. We may also interpret curvature as the **length of the acceleration vector** while travelling on the curve at $|\mathbf{v}| = 1$.

4 Multivariate Calculus

We call functions $f : D \rightarrow \mathbb{R}$, $D \subseteq \mathbb{R}^n$ a **real-valued function** on D . f is a **function of n independent variables** x_1 to x_n .

$$w = f(x_1, x_2, \dots, x_n).$$

We say w is a **dependent variable** of f .

Consider the set

$$\{(x_1, \dots, x_n, w) : \mathbf{x} = (x_1, \dots, x_n) \in D, w = f(\mathbf{x})\}$$

This set is known as the **graph** of f .

Given a constant value c , consider the set

$$\{f(\mathbf{x}) = c : \mathbf{x} \in \mathbb{R}^n\}$$

This set is known as the **level set** of f . The level set is a hypersurface for non-critical points of f . If f is continuous, the level set has to be closed.

In the case of $z = f(x, y)$, the graph is a surface of which the **projection onto the xy -plane is the domain D** , and the level sets are curves on the xy -plane. A level set can be interpreted as the **intersection between the plane $z = c$ and the graph of f** .

Interior and Boundary Points

For R a domain of f in the xy -plane

- A point is an **interior point** of R if it is the center of a small disk wholly contained in R .
- A point is a **boundary point** of R if every disk centered at the point is partially in R and partially outside R .

Limits

For n -variable function $f : D \rightarrow \mathbb{R}$, we extend the definition of **limits for multivariable functions** as follows: $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L \iff \forall \epsilon > 0 \exists \delta > 0$ such that $\forall (x,y) \in D \setminus \{(x_0,y_0)\}$,

$$\sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta \iff |f(x,y) - L| < \epsilon.$$

Suppose $L, M, k \in \mathbb{R}$ such that

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L \quad \lim_{(x,y) \rightarrow (x_0,y_0)} g(x,y) = M.$$

and $r \in \mathbb{Z}$, $s \in \mathbb{Z}^*$ with no common factors

Properties of Limits of n -Variable functions:

- $\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) + g(x,y)) = L + M$
- $\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) \cdot g(x,y)) = L \cdot M$
- $\lim_{(x,y) \rightarrow (x_0,y_0)} (kf(x,y)) = kL$
- $M \neq 0 \implies \lim_{(x,y) \rightarrow (x_0,y_0)} \left(\frac{f(x,y)}{g(x,y)} \right) = \frac{L}{M}$
- $L^{r/s} \in \mathbb{R} \implies \lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y))^{\frac{r}{s}} = L^{r/s}$

- f, g are polynomials in x, y and $g(a, b) \neq 0$, then

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$$

$$\lim_{(x,y) \rightarrow (a,b)} \frac{f(x,y)}{g(x,y)} = \frac{f(a,b)}{g(a,b)}.$$

Limits in 2 variables is more complicated as (x, y) could approach (x_0, y_0) from many directions.

We may compute by **comparison** (by “squeezing”), or conversion to other **coordinate systems**.

To show the non-existence of a limit, we then use the **two-path test**. If a function $f(x, y)$ has **different limits along two different paths** as (x, y) approaches (x_0, y_0) , then the limit $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y)$ does not exist.

4.1 Multivariate Derivatives

The **partial derivatives** of $f(x, y)$ wrt $x, f_x(x_0, y_0)$ is

$$\frac{\partial f}{\partial x} \bigg|_{x_0,y_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x},$$

provided it exists. We may interpret a partial derivative as the gradient of the tangent line to the graph $z = f(x, y_0)$ at $x = x_0$.

Similarly, we have the partial derivatives wrt y

$$\frac{\partial f}{\partial y} \bigg|_{x_0,y_0} = \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}.$$

If the first partial derivatives are defined in an open region R containing (x_0, y_0) , and are continuous at (x_0, y_0) , the **change Δz of f** by moving from (x_0, y_0) to $(x_0 + \Delta x, y_0 + \Delta y) \in R$ satisfies an equation of the form

$$\Delta z = f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y$$

where $\Delta x, \Delta y \rightarrow 0 \implies \epsilon_1, \epsilon_2 \rightarrow 0$.

We further define **higher partial derivatives** $f_{yx} = \frac{\partial^2 f}{\partial x \partial y}$. If $f(x, y)$ and its partial derivatives f_x, f_y, f_{xy}, f_{yx} are defined throughout an open region containing (a, b) and are continuous at (a, b) ,

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Directional Derivatives

The **gradient** of $f(x_1, \dots, x_n)$ at $P_0(a_1, \dots, a_n)$ is

$$\nabla f = \langle f_{x_1}, f_{x_2}, \dots, f_{x_n} \rangle.$$

$\nabla f(P_0)$ is perpendicular to the level set $f(P_0) = c$.

Properties of gradients:

For $k, l \in \mathbb{R}$, $\nabla(kf) \pm \nabla(lg) = k\nabla f \pm l\nabla g$.

$$\nabla(fg) = f\nabla g + g\nabla f, \quad \nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2}.$$

Directional derivatives generalise partial derivatives to any direction with unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$

$$\left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} = \lim_{s \rightarrow 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}$$

$$= \mathbf{u} \cdot \nabla f(x_0, y_0) = D_{\mathbf{u}} f(x_0, y_0).$$

The rate of **increase/decrease** is greatest when $\mathbf{u}$ is **in the direction of $\pm \nabla f$** . $D_{\mathbf{u}} f = \pm \sqrt{f_x^2 + f_y^2}$. The rate of change is 0 when $\mathbf{u}$ is perpendicular to ∇f , equivalently, parallel to the level set.

Tangent Space

The **tangent plane** to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$ is $f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0) = 0$. In general, the **tangent space** is defined by the specified point, and a orthogonal vector to all directional derivatives at the point. The tangent space is an **approximation of the graph**. The approximation is better in directions **parallel to the axes**.

Continuity

At $(c_0, c_1, \dots, c_n)$, the n -variable function $f : D \rightarrow \mathbb{R}$ is **continuous** if:

- $(c_0, c_1, \dots, c_n) \in D$,
- $L := \lim_{(x,\dots,x_n) \rightarrow (c_0,\dots,c_n)} f(x_0, \dots, x_n)$ exists, and
- $L = f(c_0, \dots, c_n)$.

f is **continuous** $\iff f$ is continuous $\forall (c_0, c_n) \in D$. Equivalently, f is differentiable $\implies$ continuous at the point if:

- All first order partial derivatives exist at the point,
- The tangent space is a good approximation near the point.

f is **differentiable** $\iff f$ is differentiable $\forall (c_0, \dots, c_n) \in D$.

Chain Rule

For a differentiable function $w = f(x_1, \dots, x_n)$ where each $x_i = x_i(t_1, \dots, t_m)$ is differentiable, $w(t_1, \dots, t_m) = f(x_1(t_1, \dots, t_m), \dots, x_n(t_1, \dots, t_m))$ is differentiable and

$$\frac{\partial w}{\partial t_j} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_j} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_j} + \dots + \frac{\partial f}{\partial x_n} \frac{\partial x_n}{\partial t_j}.$$

$$\left(\frac{\partial w}{\partial t_1}, \dots, \frac{\partial w}{\partial t_m}\right) = \left(\frac{\partial w}{\partial x_1}, \dots, \frac{\partial w}{\partial x_n}\right) \begin{pmatrix} \frac{\partial x_1}{\partial t_1} & \dots & \frac{\partial x_1}{\partial t_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial t_1} & \dots & \frac{\partial x_n}{\partial t_m} \end{pmatrix}$$

Extreme Values

For a n -variable function f defined on a region R containing P :

- $f(P)$ is a **local maximum value** if $f(P) > f(P')$ for all P' in an open set centred at P
- $f(P)$ is a **local minimum value** if $f(P) < f(P')$ for all P' in an open set centred at P
- P is a **local extreme point** if it is a local maximum or minimum.
- P is a **saddle point** if in every open set centred at P , there exist P' where $f(P) > f(P')$, P'' where $f(P) < f(P'')$.

An interior point is a **critical point** if one of the partial derivatives are 0 or undefined. **Extrema must occur at critical or boundary points**.

We define the **discriminant** $D = f_{xx}f_{yy} - (f_{xy})^2$ that describes the local shape of the graph. If the first and second partial derivatives are continuous in an open set centred at P and the first-order partial derivatives are 0, with the **Second Derivative Test**,

- $D < 0$ at $P \implies P$ is a **saddle point**.
- $f_{xx} < 0$ and $D > 0$ at $P \implies P$ is a **local maximum**.
- $f_{xx} > 0$ and $D > 0$ at $P \implies P$ is a **local minimum**.
- $D = 0$ at $P \implies$ The test is **inconclusive**.

Lagrange Multipliers

Within a constraint $g(x_1, x_2, \dots, x_n) = 0$, to find the maximum value of $f(x_1, x_2, \dots, x_n)$. At the maximal point P_0 , their gradients are **parallel**. For some **Lagrange multiplier** λ ,

$$\nabla f(P_0) = \lambda \nabla g(P_0).$$

For $\nabla g \neq \mathbf{0}$, P_0 is the **solution to the system of equations**

$$\begin{cases} g(x_1, x_2, \dots, x_n) = 0 \\ \nabla f(x_1, x_2, \dots, x_n) = \lambda \nabla g(x_1, x_2, \dots, x_n) \end{cases}.$$

Taylor's Expansion

Consider $f(x, y)$. We define for $a, b \in \mathbb{R}$, $n \in \mathbb{N}^*$,

$$\left(a \frac{\partial}{\partial x} + b \frac{\partial}{\partial y}\right)^n f = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \frac{\partial^n f}{\partial x^{n-k} \partial y^k}.$$

The Taylor expansion for $f(x, y)$ at the origin is

$$f(x, y) = f + xf_x + yf_y + \frac{1}{2!}(x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}) + \frac{1}{3!}(x^3 f_{xxx} + 3x^2 y f_{xxy} + 3xy^2 f_{xyy} + y^3 f_{yyy}) + \dots + \frac{1}{n!} \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right)^n f + \text{error}$$

with each partial derivative evaluated at $(0, 0)$, and a small error term (for “nice” functions and large n)

$$\text{error} = \frac{1}{(n+1)!} \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right)^{n+1} f \bigg|_{(cx, cy)}.$$

4.2 Multivariate Integrals

The **Riemann Integral** of $f(x, y)$ over a region R is the limit of partitioning R into $n \rightarrow \infty$ divisions of area $\Delta A_k \rightarrow 0$ for the **volume** of $z = f(x, y)$. That of $f(x, y, z)$ over a volume D is the limit of partitioning D each of volume ΔV_k for a **summation of f over D** . If this limit exists, f is **Riemann integral over R** .

Properties of Riemann Integrals:

- $\int_R c_1 f \pm c_2 g \, dA = c_1 \int_R f \, dA \pm c_2 \int_R g \, dA$
- $\forall (x, y) \in R, f > g \implies \int_R f \, dA > \int_R g \, dA$
- For a partition of $R = R_1 \cup R_2$,

$$\int_R f \, dA = \int_{R_1} f \, dA + \int_{R_2} f \, dA$$

By **Fubini's theorem**, we can compute the Riemann integral of f continuous over a region R , by **iterating the integration with respect to each axis**.

Applications

We find the **area/volume of the region** by setting $f = 1$ and the **average value of f over R** with $\frac{1}{\text{Area of } R} \iint_R f \, dA$.

The **surface area of the graph** consists areas $\Delta G_k = \frac{\Delta A_k}{\mathbf{n} \cdot \mathbf{k}}$, where

$$\mathbf{n} \cdot \mathbf{k} = \frac{\langle -f_x, -f_y, 1 \rangle}{\sqrt{1 + f_x^2 + f_y^2}} \cdot \langle 0, 0, 1 \rangle \implies$$

$$\text{Area} = \iint_R \sqrt{1 + f_x^2 + f_y^2} \, dxdy.$$

For an object occupying D in 3-dim space with density $\delta(x, y, z)$,

- Mass** $M = \iiint_D \delta(x, y, z) \, dV$
- Moment about yz -plane $M_{yz} = \iiint_D x \delta(x, y, z) \, dV$
- Moment about xz -plane $M_{xz} = \iiint_D y \delta(x, y, z) \, dV$
- Moment about xy -plane $M_{xy} = \iiint_D z \delta(x, y, z) \, dV$
- Center of mass:** $\bar{x} = \frac{M_{yz}}{M}, \bar{y} = \frac{M_{xz}}{M}, \bar{z} = \frac{M_{xy}}{M}$

Substitutions

In **polar/cylindrical coordinates**, $\Delta A_k = r_k \Delta r_k \Delta \theta_k$, and in **spherical coordinates**, $\Delta V = \rho^2 \sin \phi \, \Delta \rho \Delta \phi \Delta \theta$.

$$dA = r dr d\theta, \quad dV = r dr d\theta dz = \rho^2 \sin \phi \, d\rho d\phi d\theta.$$

To perform substitutions for a function F ,

$$x = g(u, v, w), \quad y = h(u, v, w), \quad z = k(u, v, w)$$

$$H(u, v, w) = F(g(u, v, w), h(u, v, w), k(u, v, w))$$

The map from the uvw -coordinate space to the domain of integration $T : (u, v, w) \mapsto \langle g(u, v, w), h(u, v, w), k(u, v, w) \rangle$ must be a bijection, and the **Jacobian** yields the multiple

$$J(u, v, w) = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix}$$

$$\begin{aligned} & \iiint_R F(x, y, z) \, dx \, dy \, dz \\ &= \iiint_D H(u, v, w) |J(u, v, w)| \, du \, dv \, dw \end{aligned}$$

5 Vector Fields

A **vector field** on $\mathbb{R}^n$ is a function $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$. A **gradient field** is the field of gradients

$\nabla f(x, y, z) = \langle f_x(x, y, z), f_y(x, y, z), f_z(x, y, z) \rangle$, where f is the potential function of the gradient field. The **line integral** along segment C is given by

$$\int_C f ds = \int_b^a f(\mathbf{r}(t)) |\mathbf{v}(t)| \, dt, \quad \text{for some smooth } \mathbf{r}$$

The line integral is dependent on the path C , but not on the motion $\mathbf{r}, \mathbf{v}$. It could be interpreted as the **area under the path**.

Properties of Vector Fields

Let $\mathbf{F}$ be a vector field on an open connected domain D .

If $\forall A, B \in D$, $\int_A^B \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A)$ is the same over all paths from A to B , the integral is **path independent** in D and $\mathbf{F}$ is **conservative** on D . If $\mathbf{F} = \nabla f$ for some scalar function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, f is a potential function for $\mathbf{F}$.

$\mathbf{F}$ is a gradient field $\iff \mathbf{F}$ is a conservative field

$$\begin{aligned} &\iff \oint_C \mathbf{F} \cdot d\mathbf{r} = 0 \\ &\iff \operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \langle 0, 0, 0 \rangle \end{aligned}$$

Applications

The **work done** by a force $\mathbf{F}$ over a smooth curve $\mathbf{r}(t)$ is

$$W = \int_{t=a}^{t=b} \mathbf{F} \cdot \mathbf{T} \, ds = \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt.$$

The work integral is independent of the motion $\mathbf{r}$, but is path dependent.

The **flow** along a curve for some smooth $\mathbf{r}$ in a continuous velocity field $\mathbf{F}$ is

$$\text{Flow} = \int_a^b \mathbf{F} \cdot \mathbf{T} \, ds.$$

If the curve is a closed loop, the flow integral is the **circulation about the curve**. Mathematically, it is identical to the work done.

Green's Theorem

A **closed curve** is one whose starting point = end point. A **simple curve** is one that does not cross itself. A **directed/oriented** closed curve is one with a direction. An oriented simple closed curve moving in an **anti-clockwise direction** has a **positive** orientation, and negative orientation otherwise.

For a positive oriented closed curve C , R its enclosed region, and $\mathbf{F}(x, y) = \langle M(x, y), N(x, y) \rangle$ a differentiable vector field on the xy -plane

$$\begin{aligned} \oint_C M \, dx + N \, dy &= \iint_R \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \, dx \, dy. \\ \oint_C M \, dy - N \, dx &= \iint_R \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \, dx \, dy. \end{aligned}$$

For vector fields undefined at the origin, we consider a curve C_h clockwise around the origin of radius h .

Surface Integrals

The area of surface S $f(x, y, z) = c$ over a closed and bounded plane region R is, where $\mathbf{p}$ is a the unit vector normal to R ,

$$\text{Surface Area} = \iint_R \frac{|\nabla f|}{|\nabla f \cdot \mathbf{p}|} \, dA.$$

For a continuous g defined at points of S , the surface integral is

$$\iint_S g d\sigma = \iint_R g(x, y, z) \frac{|\nabla f|}{|\nabla f \cdot \mathbf{p}|} \, dA.$$

Intuitively, an **orientation** of a surface is defining a unit normal vector $\mathbf{n}(p)$ for every point p on a surface, such that the vector changes **continuously** wrt. p . A surface with an orientation is orientable. A connected surface has either 0 or 2 orientations, and closed surfaces are orientable.

The **flux** of a vector field $\mathbf{F}$ across an oriented surface S in the direction of $\mathbf{n}$ is

$$\text{Flux} = \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F}(x, y, z) \cdot \mathbf{n} \, d\sigma.$$

Intuitively, it is the "amount of flowing" across a surface. For a surface S defined by $g(x, y, z) = 0$,

$$\text{Flux} = \iint_R \mathbf{F} \cdot \frac{\nabla g}{|\nabla g \cdot \langle 0, 0, 1 \rangle|} \, dx \, dy$$

Stokes' Theorem

We define the curl the "amount of spinning" in a vector field. We use the right-hand rule for the direction of the curl.

$$\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \left\langle \frac{\partial P}{\partial y} - \frac{\partial N}{\partial z}, \frac{\partial M}{\partial z} - \frac{\partial P}{\partial x}, \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right\rangle.$$

By Green's theorem, $\oint_C \mathbf{C} \cdot d\mathbf{r} = \iint_R \operatorname{curl} \mathbf{F} \cdot \mathbf{k} \, dA$ The circulation (flow integral) a vector field $\mathbf{F} = \langle M, N, P \rangle$ around the boundary C of an oriented surface S with $\mathbf{n}$ in the counter-clockwise direction

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} \, d\sigma.$$

Divergence Theorem

We define the divergence the "amount of expansion/production" in a vector field.

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}.$$

The field is incompressible if $\operatorname{div} = 0$ at all points. The theorem states

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iiint_D \operatorname{div} \mathbf{F} \, dV$$